

Question 1

0.6818 points

✓ Saved

The joint P.d.f of the random variable X and Y defined as follows:

$$f(x, y) = 9x^2y^2, \quad 0 \leq y \leq x \leq 1$$

0

otherwise

Solve questions from 1 to 10

$f(x) =$

☐ A. $4x^3$, $0 \leq x \leq 1$

☒ B. $3x^5$, $0 \leq x \leq 1$

☐ C. $8x^5$, $-1 \leq x \leq 1$

☐ D. $4x^3$, $-2 \leq x \leq 2$

$$\begin{aligned} & \int_0^x 9x^2y^2 dy \\ &= 9x^2 \int_0^x y^2 dy \\ &= 9x^2 \left[\frac{y^3}{3} \right]_0^x \\ &= 9x^2 \left[\frac{x^3}{3} \right] = 3x^5 \quad 0 \leq x \leq 1 \end{aligned}$$

→ ⚠ Moving to another question will save this response.

Question 2

$$f(y) =$$

- ☐ A. $3y^3(1-y^2)$, $0 \leq y \leq 1$
- ☐ B. $4y(1-y^2)$, $-1 \leq y \leq 1$
- ☒ C. $3y^2(1-y^3)$, $0 \leq y \leq 1$
- ☐ D. $4y(1-y^2)$, $-2 \leq y \leq 2$

$$\begin{aligned} & \int_y^1 9x^2y^2 \, dx \\ &= 9y^2 \int_y^1 x^2 \, dx \\ &= 9y^2 \left[\frac{x^3}{3} \right]_y^1 \\ &= 9y^2 \left[\frac{1}{3} - \frac{y^3}{3} \right] \\ &= 9y^2 \left[\frac{1-y^3}{3} \right] \\ &= 3y^2(1-y^3) \quad 0 \leq y \leq 1 \end{aligned}$$

→ ⚠ Moving to another question will save this response.

Question 3

E (x) = $\int_0^1 x \cdot 3x^5 dx$

☐ A. 5/4

$$= 3 \int_0^1 x^6 dx$$

☐ B. 4/5

$$= 3 \left[\frac{x^7}{7} \right]_0^1$$

☐ C. 2/3

$$= 3 \left[\frac{1}{7} \right]$$

☒ D. 3/7

$$= \frac{3}{7}$$

Question 4

$$E(y) = \int_0^1 y \cdot 3y^2 [1 - y^3] dy$$

☐ A. 5/4

$$= \int_0^1 3y^3 [1 - y^3] dy$$

☒ B. 9/28

$$= 3 \int_0^1 y^3 - y^6 dy$$

☐ C. 4/5

$$= 3 \left[\frac{y^4}{4} - \frac{y^7}{7} \right]_0^1$$

☐ D. 1/2

$$= 3 \left[\frac{1}{4} - \frac{1}{7} \right]$$

$$= 3 \left[\frac{3}{28} \right]$$

$$= \frac{9}{28}$$

→ ⚠ Moving to another question will save this response.

Question 4

$$E(y) =$$

- ☐ A. $5/4$
- ☒ B. $9/28$
- ☐ C. $4/5$
- ☐ D. $1/2$

So

Question 5

$$E(x^2) =$$

☐ A. $5/4$

☒ B. $3/8$

☐ C. $2/3$

☐ D. $1/2$

$$\begin{aligned} & \int_0^1 x^2 3x^3 dx \\ &= 3 \int_0^1 x^7 dx \\ &= 3 \left[\frac{x^8}{8} \right]_0^1 \\ &= 3 \left(\frac{1}{8} \right) \\ &= 3/8 \end{aligned}$$

→ ⚠ Moving to another question will save this response.

Question 6

$$E(y^2) =$$

☐ A. 7/40

☒ B. 9/40

☐ C. 2/30

☐ D. 4/50

$$\int_0^1 y^2 3y^2 [1-y^3] dy$$

$$= 3 \int_0^1 y^4 [1-y^3] dy$$

$$= 3 \int_0^1 y^4 - y^7 dy$$

$$= 3 \left[\frac{y^5}{5} - \frac{y^8}{8} \right]_0^1$$

$$= 3 \left[\frac{1}{5} - \frac{1}{8} \right] = 3 \left[\frac{3}{40} \right] = \frac{9}{40}$$



Moving to another question will save this response.

Question 7

$V(x) =$

☐ A. 0.345

☐ B. 0.755

☐ C. 0.897

☒ D. 0.191

$$E(x^2) - [E(x)]^2$$

$$= \frac{3}{8} - \left(\frac{3}{7}\right)^2$$

$$= \frac{3}{8} - \frac{9}{49}$$

$$= \frac{75}{392} = 0.1913$$

Question 8

$$V(y) =$$

☐ A. 0.654

☐ B. 0.981

☒ C. 0.122

☐ D. $\frac{1}{2}$

$$E(y^2) - [E(y)]^2$$

$$\frac{9}{40} - \left(\frac{9}{28}\right)^2$$

$$= \frac{9}{40} - \frac{81}{784}$$

$$= 0.122$$

Question 9

$$\text{Cov}(x, y) =$$

☒ A. 0.143

☐ B. 2/3

☐ C. 1/2

☐ D. 0.445

$$E(xy) - E(x)E(y)$$

$$E(xy) = \int_0^1 \int_0^1 xy \cdot 9x^2y^2 \, dx \, dy$$

$$= \int_0^1 9y^3 \int_0^1 x^3 \, dx \, dy$$

$$= \int_0^1 9y^3 \left[\frac{x^4}{4} \right]_0^1 \, dy$$

$$= \int_0^1 9y^3 \left[\frac{1}{4} - \frac{0}{4} \right] \, dy$$

$$= \int_0^1 9y^3 \left[\frac{1-y^4}{4} \right] \, dy$$

$$= \frac{9}{4} \int_0^1 y^3 - y^7 \, dy$$

$$= \frac{9}{4} \left[\frac{y^4}{4} - \frac{y^8}{8} \right]_0^1$$

$$= \frac{9}{4} \left[\frac{1}{4} - \frac{1}{8} \right]$$

$$= \frac{9}{4} \left[\frac{1}{8} \right]$$

$$= \frac{9}{4} \left[\frac{1}{8} \right] = \frac{9}{32}$$

$$= \frac{9}{32} - \frac{3}{7} \left(\frac{9}{28} \right)$$

$$= \frac{9}{32} - \frac{27}{196}$$

$$= 0.14355$$

Question 10

ρ_{xy}

$$\frac{0.143}{\sqrt{0.1913(0.122)}}$$

$$= 0.937$$

☐ A. 0.658

☐ B. 0.143

☐ C. 4/5

☒ D. 0.937

Question 11

0.6818 points

A fair coin is tossed three times. Let x denotes the number of heads obtained in the first two tosses and y denotes the number of heads obtained in the last toss. Solve questions from 11 to 18

The joint PMF of x and y is:

☐ A.

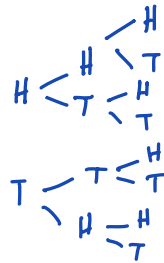
y	0	1	2
x			
0	1/32	5/32	6/32
1	8/32	7/32	5/32
2	10/32	11/32	11/32
3	0	2/32	7/32

☐ B.

y	0	1	2	3
x				
0	1/32	5/32	6/32	7/32
1	8/32	7/32	5/32	10/32
2	10/32	11/32	11/32	4/32
3	0	2/32	7/32	9/32

☒ C.

y	0	1
x		
0	1/8	1/8
1	2/8	2/8
2	1/8	1/8



HHH
HHT
HTH
HTT
TTH
TTT
THT
TTT

x
2
2
1
1
0
0
1
1

y
1
0
1
0
1
0
1
0

y	0	1	$P(x)$
0	1/8	1/8	2/8
1	2/8	2/8	4/8
2	1/8	1/8	2/8
$P(y)$	4/8	4/8	1

The marginal PMF of x is: *from joint pdf*



A.

x	0	1	2
f(x)	2/8	4/8	2/8



Question 13

The marginal PMF of y is: *From joint pdf*

☐ A.

y	0	1
f(y)	1/4	3/4

☐ B.

y	0	1	2
f(y)	3/8	2/8	8/32

☒ C.

y	0	1
f(y)	4/8	4/8



☐ D.

y	0	1	2
f(y)	5/32	7/32	8/32



Moving to another question will save this response.

Question 14

$E(x) =$

$$0\left(\frac{2}{8}\right) + 1\left(\frac{4}{8}\right) + 2\left(\frac{2}{8}\right)$$

☐ A. $1/2$

$$= \frac{4}{8} + \left(\frac{4}{8}\right) = 1$$

☒ B. 1

☐ C. $4/5$

☐ D. $2/3$



Moving to another question will save this response.

Question 15

$E(y) =$

$$0\left(\frac{4}{8}\right) + 1\left(\frac{4}{8}\right)$$

☒ A. $1/2$

$$= \frac{1}{2}$$

☐ B. $4/5$

☐ C. $5/4$

☐ D. $2/3$

Question 16

$$E(xy) =$$

☐ A. 0

☒ B. 1/2

☐ C. 1/4

☐ D. 1/3

$$E(x) \cdot E(y)$$

$$= 1 \left(\frac{1}{2} \right)$$

$$= \frac{1}{2}$$

Question 17

$$\sigma_{xy} =$$

$$E(xy) - E(x)E(y)$$

$$\frac{1}{2} - (1)\left(\frac{1}{2}\right) = 0$$

☒ A. 0

☐ B. 1/2

☐ C. 1/4

☐ D. 1/3

Question 18

$$\rho_{xy} =$$

if $\text{cov}_{xy} = 0 \rightarrow \rho_{xy} = 0$

☐ A. -1

☒ B. 0

☐ C. 1

☐ D. 1/2

Question 19

The joint CDF of two continuous random variables x and y is given by:

$$F(x, y) = 2 - e^{-2ax} - e^{-2by} + e^{-(ax+by)} \quad x, y \geq 0$$

Solve questions form 19 to 22

The joint PDF of x and y is:

- ☒ A. $a b e^{-(ax+by)}$, $0 < x < \infty$, $0 < y < \infty$
- ☐ B. $2 a b e^{-(ax+by)}$, $0 < x < \infty$, $0 < y < \infty$
- ☐ C. $a b e^{(ax+by)}$, $0 < x < \infty$, $0 < y < \infty$
- ☐ D. $a b e^{-2(ax+by)}$, $0 < x < \infty$, $0 < y < \infty$

$$f(x, y) = \frac{d^2}{dx dy} [2 - e^{-2ax} - e^{-2by} + e^{-(ax+by)}]$$

$$= \frac{d}{dx} [2b e^{-2by} - b e^{-(ax+by)}]$$

$$= a b e^{-(ax+by)}$$

Question 20

The marginal PDF of x is:

☒ A. $a e^{-ax}$, $0 < x < \infty$

☐ B. $a b e^{-2(ax+by)}$, $0 < x < \infty$

☐ C. $2 a b e^{-(ax+by)}$, $0 < x < \infty$

☐ D. $a e^{-(ax+by)}$, $0 < x < \infty$

$$\int_0^{\infty} a b e^{-(ax+by)} dy$$

$$a b e^{-ax} \int_0^{\infty} e^{-by} dy$$
$$= \frac{a b e^{-ax}}{-b} [e^{-by}]_0^{\infty}$$

$$= -a e^{-ax} [e^{-b(\infty)} - e^{-b(0)}]$$

$$= -a e^{-ax} [-1]$$
$$= a e^{-ax}$$

Question 21

The marginal PDF of y is:

- ☒ A. $b e^{-by}$, $0 < y < \infty$
- ☐ B. $a b e^{-(ax+by)}$, $0 < y < \infty$
- ☐ C. $a e^{ax}$, $0 < y < \infty$
- ☐ D. $2 a b e^{-(ax+by)}$, $0 < y < \infty$

Question 22

$$F(0.75, 0.75) =$$

- ☒ A. $2 - e^{-1.5a} - e^{-1.5b} + e^{-0.75a - 0.75b}$
- ☐ B. $2 - e^{-a} - e^{-b} + e^{0.75a + 0.75b}$
- ☐ C. $a b e^{-2(ax+by)}$
- ☐ D. $1 - e^{-1.5a} - e^{-1.5b} + e^{-0.75a - 0.75b}$